Term 1 answer key

Formulas printed for students in every question. Lengths, areas and volumes are rounded only where the question says so (1 decimal place, the nearest degree or the nearest cent).

WEEK	SESSION	Q1 A Triangles & similarity	Q2 B Area & volume	Q3 C Percentages & money	Q4 D Indices, formulas & rates	Q5 E Graphs, data & chance
1	1	9.1 cm	120 cm^3	$3600	$20 and $70	3/8
	2	45°	160 m^2	25% loss	y^5	7
	3	26.9 cm	153.9 m^2	2520	6	0.85
2	1	34 cm	36 L	$450	3.6×10^6	0.36 or 9/25
	2	16 cm	408.4 cm^2	$200	2	3
	3	6 cm	8.4 cm^2	$6345	20 km/h	6
3	1	24/25	62.8 cm^3	$120	x^8	1/3
	2	24 cm	6.3 cm	$3200	2 750 000	0.65
	3	6.9 cm	104.7 cm^2	$63	5 hours	2
4	1	25 cm	50.3 cm^2	$360	1	13
	2	8/15	8 cm	$3760	$45 and $75	1
	3	6 cm	12.6 cm	$1200	26	60
5	1	15.7 cm	72 L	50% profit	32 000	1/2
	2	72°	90 cm^2	$1218	m^8	14
	3	24/7	138.2 m^2	$640	90 km/h	3/4
6	1	5 cm	113.1 cm^3	$80	$300 and $200	0.42 or 21/50
	2	6.2 cm	92 cm^2	20% loss	$1/5^2$ = 1/25	2
	3	23°	108 cm^3	$13 000	5	gradient −4, y-intercept 1
7	1	7 cm	28.3 cm^2	$4888	4 km/h	50
	2	5.9 cm	200 L	$2200	5^6	1/5
	3	17 cm	452.4 cm^3	$300	18 and 30	7
8	1	8.8 cm	251.3 cm^2	$920	5	2
	2	24 cm	28.3 m	$400	$1/3^4$ = 1/81	11
	3	66°	314.2 cm^2	200 kg	8.2×10^2	3/5
9	1	12/13	56.5 m^3	$9600	$60 and $90	1
	2	13 cm	126 cm^2	$4089	10	0.3 or 3/10
	3	9.4 cm	113.1 cm^2	$1500	610 000	1/3
10	1	10 cm	5 L	$360	70 km/h	5/6
	2	10.0 cm	150 cm^3	90	1/100	4
	3	12 m	879.6 cm^2	25% profit	$4n^4$	4

NSW MATHEMATICS K–10 (2022) OUTCOMES PRACTISED · STAGE 5, WITH STAGE 4 REVISION · MAO-WM-01 THROUGHOUT

A MA4-PYT-C-01 Pythagoras' theorem; MA5-TRG-C-01 trigonometric ratios (sides and angles); MA5-TRG-C-02 angles of elevation and depression (Terms 3–4); MA5-GEO-C-01 similar figures

B MA4-ARE-C-01 area of circles and sectors*; MA4-LEN-C-01 arc length*; MA5-ARE-C-01 surface area of prisms and cylinders; MA5-VOL-C-01 volume of prisms and cylinders, capacity; MA5-VOL-P-01 (Path, Term 4 only) volume of pyramids and cones

C MA5-FIN-C-01 simple interest, earning money (wages, overtime, salary, commission, leave loading); MA4-FRC-C-01 percentage increase and decrease, original amount, percentage profit and loss*

D MA5-IND-C-01 index laws with coefficients, zero and negative indices (numerical bases); MA5-MAG-C-01 scientific notation; MA4-ALG-C-01 / MA5-EQU-C-01 substituting into formulas and solving*; MA4-RAT-C-01 ratios and rates (speed)

E MA5-LIN-C-01 gradient and $y = mx + c$; MA4-DAT-C-02 mean, including frequency tables*; MA4-PRO-C-01 / MA5-PRO-C-01 probability, complementary events, relative and expected frequency*

* the code is right for the topic, but where this content sits within it could not be checked against the NESA website.